

Gene Pulser® Electroprotocols

* We recommend adapting this protocol to use the Gene Pulser electroporation buffer (catalog #165-2676, 165-2677), which increases cell viability and transfection efficiency in mammalian cell lines.

Cell Type	Mammalian, adherent	Molecules Electroporated	DNA: linearized, (pMSG-derivative)
Species Used	Rat, D202CC, hepatoma; Human, TCCSUP (epithelial-like) bladder carcinoma		

Before the Pulse

Cell Growth Medium	MEM, 10% Fetal Calf Serum, (+aminopterin, MPA, hypoxanthin, xanthine, thymidine) (GIBCO/BRL, Sigma)	Growth Phase at Harvest	Log phase
		Pre-pulse Incubation	10 min. on ice, in Phosphate Buffered Sucrose
Wash Solution	Phosphate Buffered Sucrose		

The Pulse

Instruments Used Gene Pulser® apparatus

Electroporation Temperature	Room temperature		
Electroporation Medium*	Phosphate Buffered Sucrose (272 mM sucrose, 7 mM potassium phosphate, pH 7.4, 1 mM MgCl ₂)	Cuvette Gap	0.4 cm
Cell Density	1 to 50 x 10 ⁵ (5) cells / ml	Voltage	0.20 to 0.35 kV
Volume of Cells	0.4 ml	Field Strength	0.5 to 0.875 kV/cm
DNA Concentration	Not given	Capacitor	25 µF
DNA Resuspension Buffer	Not given	Resistor	(Pulse Controller) Ω none
Volume of DNA	5 to 20 µg /ml	Time Constant	7 to 12 msec
After the Pulse			
Outgrowth Medium	As above		

Relevant Publications and/or Comments

Note: exponential values designated in parentheses.

Outgrowth Temperature	Not given
Length of Incubation	Stable transfectants
Selection Method or Assay Used	MPA
Electroporation Efficiency	1 to 10 x 10 ⁶ (6) cells
Per Cent Survival	Not given

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